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Classes

Classes composition:

Input File **Structure of the input file:**

Mesh points: Number of points in X, Y, Z directions

Example: `50 50 50` $\Rightarrow Nx = 50, Ny = 50, Nz = 50$

Time parameters:

`dt` - Time step size, e.g., `0.01`

`T` - Total simulation time, e.g., `10.0`

Spatial step sizes:

`dx dy dz` - Grid spacing in X, Y, Z, e.g., `0.5 0.5 0.5`

Initial conditions files:

`u0_init_file` - File with initial velocity values

`p0_init_file` - File with initial pressure values

`k_file` - File with permeability values

Boundary conditions analysis

[shrink]Velocity Boundary Conditions For each of the three steps of the algorithm for momentum equation, the Di
when solving for the component parallel to the decomposition direction:

when solving for a component perpendicular to the decomposition direction:

Boundary Conditions: Indexing Pattern

We found a structural pattern in our staggered grid that simplifies boundary condition (BC) logic. Let x_0 be the s
0.6

Normal Component: i_1 is **even** AND i_2 is **even**.

Tangential Component: i_1 is **odd** OR i_2 is **odd**.

0.4 [width=keepaspectratio]img/block$_{s}olver.png$ This pattern holds for all **three velocity components** and allows fo

Future workflow Code Validation

Compute error

Compute residual

Convergence study

Parallelization Strategy

Vectorization of the sequential code

Parallelization of the code